Lesson 5:
Automated evaluators

(and, automating automated evaluators)

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Last time / today

Last time

Read traces and define failure modes.

Today

Build evaluators and check their decisions against human labels.

Code checks for rules we can compute

e.g., check every called tool against the allowed tool list.

LLM judges for criteria requiring interpretation

e.g., assess whether a refund reply gives enough information without unnecessary detail.

An example criterion for an LLM judge

Criterion: The final message includes the information the user needs for their next decision, without irrelevant or excessive detail.

Grice's maxims

Quantity: Include what is needed, and no more.

Quality: Say what is true and supported by evidence.

Relation: Stay relevant to the request.

Manner: Avoid ambiguity and present information in order.

Cookbook return, trace excerpt

User

I got a cookbook from Paper Lantern Press and some of the pages were stuck together. Can I return it?

Tool call

```
get_order({"order_id": 2103})
```

Tool result, selected fields

```
{"order_id": 2103, "total_usd": 255.75}
```

Assistant, final reply excerpt

If you'd like, I can help start the refund/return for order #2103 due to the stuck-together pages. Since this is a higher-value order, it may be routed for human approval after submission.

Multiple issues: no refund amount, no explanation of "higher-value," and unclear whether human approval is required.

Developing an LLM judge

Given: 100 traces, each labeled Pass or Fail by a human for one criterion.
For example, 50 Pass and 50 Fail.

Step 1: Split the traces into training, development, and test sets.

Training: 20

10 Pass / 10 Fail

Select examples
for the prompt.

Development: 40

20 Pass / 20 Fail

Run, inspect
disagreements, revise.

Test: 40

20 Pass / 20 Fail

Freeze prompt and model.
Test once.

Common mistake: leakage
Using dev/test examples in the prompt, or tuning on test results.

Live: build and refine a judge

Cartwheel final messages, with a coding agent

write-judge-prompt
validate-evaluator

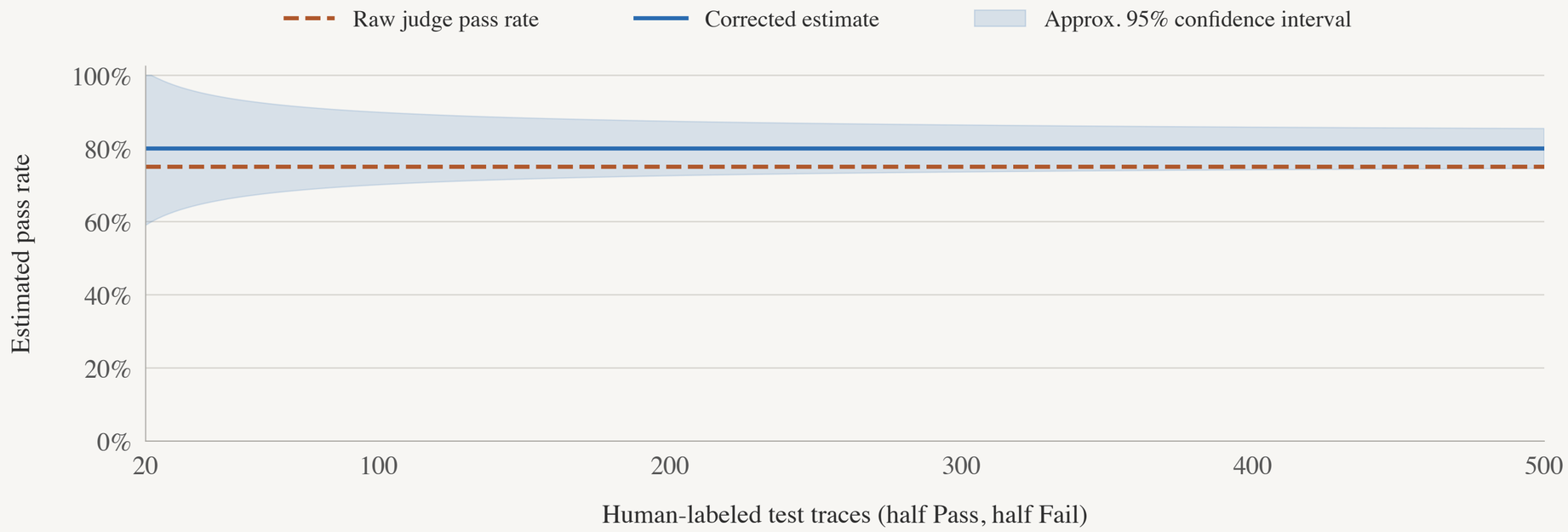
Whiteboard: measuring the judge

Confusion matrix and TPR / TNR
Corrected prevalence and uncertainty

Positive = Pass Negative = Fail

What happens when we label more traces?

Illustration: 80% true pass rate, 1,000 target traces
TPR 90%, TNR 85%

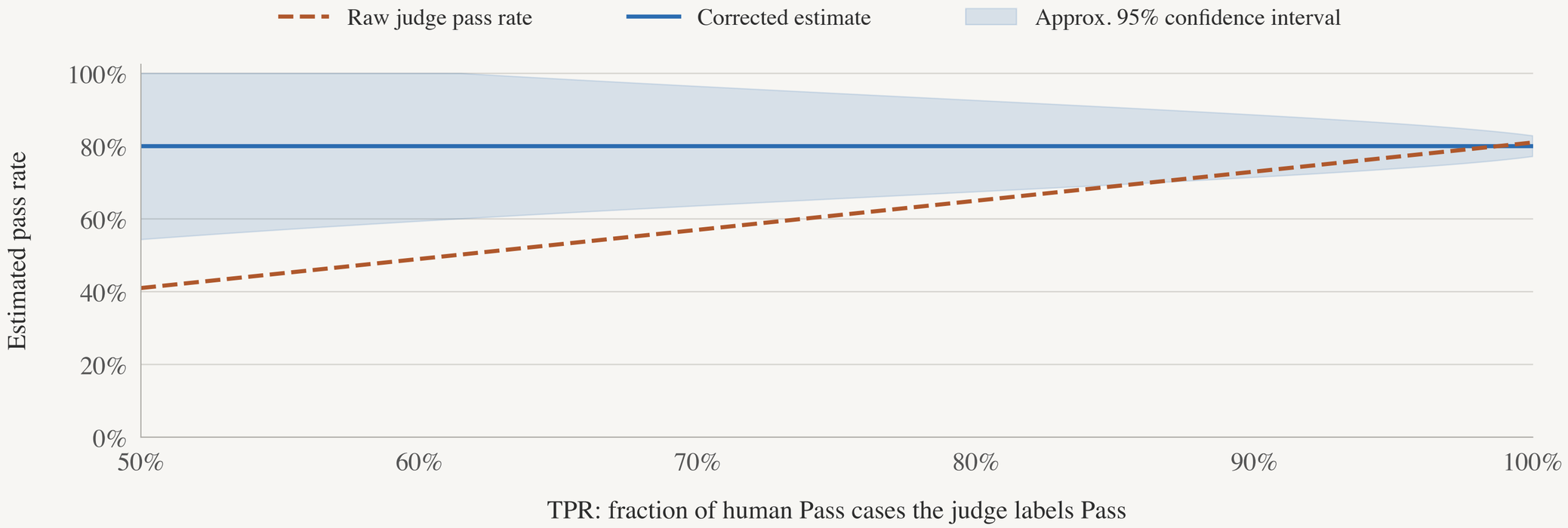


The shaded band is a **95% confidence interval** for the true pass rate. Across repeated samples, about 95% of intervals calculated the same way would contain the true rate.

Labeling more data narrows the confidence interval.

What happens when the judge misses passes?

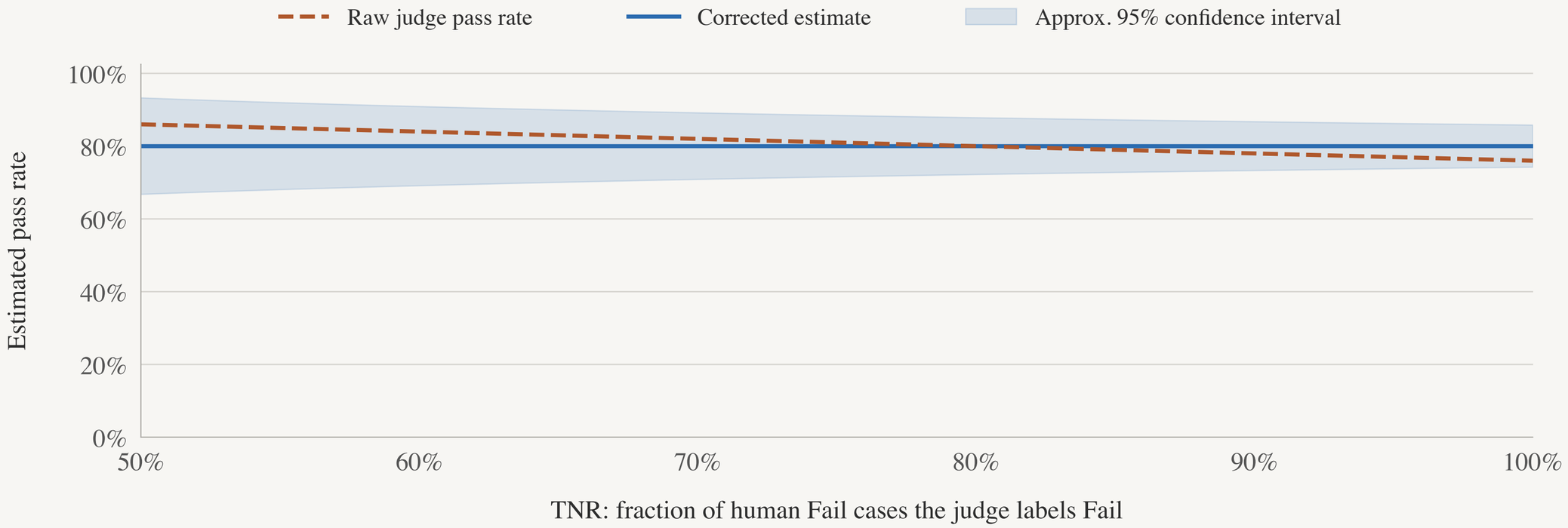
Illustration: 80% true pass rate, 1,000 target traces
100 test labels (50 Pass / 50 Fail), TNR 95%



Low TPR undercounts passes and makes the correction less precise.

What happens when the judge misses failures?

Illustration: 80% true pass rate, 1,000 target traces
100 test labels (50 Pass / 50 Fail), TPR 95%



Low TNR counts failures as passes.

Homework 5 and office hours

- **A** Choose one Homework 4 failure mode to evaluate with an LLM judge.
- **B** Split your labeled traces into training, development, and test sets.
- **C** Use write-judge-prompt to draft your judge. Use validate-evaluator to review errors.
- **D** Test each final prompt against your labels. Report TPR, TNR, and disagreements.

When	What
Sat 9/19, 12:15 PM PT	Office hours
Wed 9/23, 8:00 AM PT	Office hours, following the lecture

Schedule and joining links on Maven

Homework 5 instructions (two more judges optional)

Course reader